ML PROJECT · CHRISTIE'S AUCTIONS 1989–2026



$450,312,500

Leonardo da Vinci · 2017

What is a painting worth?

Predicting auction prices from the catalogue alone

Dmytro Lapko · Mykhailo Ostapchuk

TASK

Predict the sale price — without Christie's estimate



$195,040,000

Andy Warhol · 2022

artist · size · medium · year painted · signed · sale date & type

→ $?

Prices range from **$37** to **$450,312,500** · tested on future sales (2022+)

DATA

We built our own dataset from Christie's



$179,365,000

Pablo Picasso · 2015

560k
lots

121k
artists

8,249
sales, 1989–2026

34k
images analysed

Scrape → Parse text → 64-byte binary records · 80 MB → Model

APPROACH

35k USD paintings, models from simple to strong



$121,160,000

René Magritte · 2024

Median → Artist average → Ridge → Random forest → **Gradient boosting**

Train + time-ordered CV · 2010–2021 · 27,636 lots | **Test 2022+**

Every model with and without the artist · also tested on unseen artists

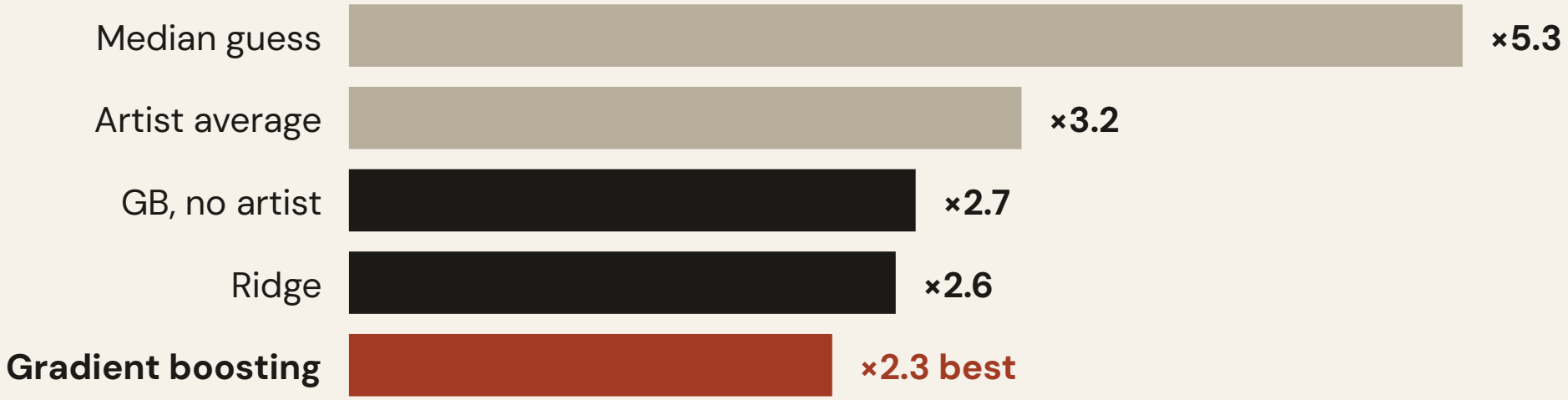
RESULTS · 2022+ TEST

Typical error: from ×5.3 down to ×2.3



$117,180,000

Vincent van Gogh · 2022



Artists seen before ×2.2 · artists new to the model ×3.0

WHAT WE TRIED

What worked, what didn’t



$91,875,000

Edward Hopper · 2018

Worked

Gradient boosting

Artist’s auction history

Sale context: evening sale, lot position

Didn’t

Colour & image statistics

Random forest, plain linear

Data traps

Clocks & documents filed as paintings

One artist, several names

CONCLUSION

Reputation and placement matter more than the painting itself



$90,312,500

David Hockney · 2018

TEAM

Dmytro Lapko — data & scraping

Mykhailo Ostapchuk — modelling & analysis

TOOLS

Python · scikit-learn · Jupyter · GitHub · Claude Code

TIME

~2 weeks · 21 Oct – 4 Nov